De Mathematics Competitions

3rd Annual

DIME I

De Invitational Mathematics Examination I

Friday, September 25, 2026



INSTRUCTIONS

1. DO NOT TURN THE PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 15-question competition. All answers are integers ranging from 000 to 999, inclusive. Make sure each of your answers is 3 digits, even if the first digit is zero.
3. Mark your answer to each problem on the answer sheet by completely filling the circle with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet and appear sufficiently dark will be scored.
4. SCORING: You will receive 1 point for each correct answer, 0 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. Prohibited materials include calculators, smartwatches, phones, computing devices, protractors, and graph paper.
6. Figures are not necessarily drawn to scale.
7. You will have 3 hours to complete the competition once your competition manager tells you to begin, though all work must be submitted by Friday, January 29, by 11:59 pm ET regardless of starting time.
8. You may take the DIME as many times as you want. Taking both the DIME I and II is highly encouraged.

The problems and solutions for this DIME were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

The DMC Committee reserves the right to disqualify scores if it determines that the rules were not followed. Participants are subject to the DMC Policies on Competition Integrity, which can be found on our site.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

Problem 1:
Find the sum of all three-digit positive integers $\underline{a}\,\underline{b}\,\underline{c}$ that satisfy

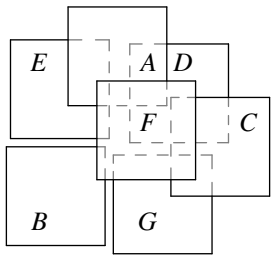
$$\underline{a}\,\underline{b}\,\underline{c} = (b+c)^a,$$

where a, b, and c are (not necessarily distinct) digits, and a is nonzero.

Problem 2:
Ayako's bakery is open for the next seven days, and exactly one of its four bakers, Billy, Carlos, Dahlia, and Ehuang, works each day. The numbers of days they each work must be divisible by 2, 3, 4, and 5, respectively (possibly zero). Find the number of ways to assign the bakers to the seven days.

Problem 3:
Let N be a three-digit positive integer with distinct nonzero digits. Suppose that exactly one of its digits is a divisor of 48, exactly one of its digits is a divisor of 56, and exactly one of its digits is a divisor of 60. Find the difference between the greatest and least possible values of N.

Problem 4:
Katherine places 7 congruent square tiles, labeled A through G, on a table one at a time. Each tile is placed flat on the table, possibly covering parts of tiles placed earlier. The figure below shows a top-down view of the table, with each tile's outline solid where the tile is still visible and dashed where it is under a later tile.



Find the number of orders in which Katherine could have placed the tiles.

Problem 5:
Let $ABCD$ be a square with side length 7, and let E be a point on $\overline{BC}$ such that $CE = 4$. Suppose that G is a point on $\overline{AD}$ and $F \neq C$ is a point on $\overline{CG}$ such that $AEFG$ is an isosceles trapezoid. The area of $AEFG$ is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Problem 6:
Let a, b, and c be real numbers greater than 1 such that the expression

$$2\sqrt{\log_2 a \cdot \log_2 b} + 6\sqrt{\log_2 b \cdot \log_2 c} + 3\sqrt{\log_2 c \cdot \log_2 a}$$

is equal to each of $144 - \log_2 a$, $126 - 4\log_2 b$, and $112 - 9\log_2 c$. Find $\log_2(abc)$.

Problem 7:
Let a, b, and c be consecutive positive integers such that

$$500 \leq \gcd(a! + b! + c!,\ abc) \leq 1000.$$

Find the sum of all possible values of $a+b+c$.

Problem 8:
A list a_1, a_2, a_3, a_4, a_5 of positive integers at most 12 has the property that for every integer n, at least one of

$$\frac{n-a_1}{2}, \frac{n-a_2}{3}, \frac{n-a_3}{4}, \frac{n-a_4}{6}, \frac{n-a_5}{12}$$

is an integer. Find the remainder when the number of such lists is divided by 1000.

Problem 9:
In $\triangle ABC$ with $BC = 14$, let M and N be the midpoints of $\overline{AB}$ and $\overline{AC}$. Line AB meets the circumcircle of $\triangle BCN$ at $P \neq B$, with B between A and P, and line AC meets the circumcircle of $\triangle BCM$ at $Q \neq C$, with Q between A and N. Suppose that $PM = 10$ and $PQ = 13$. Find the square of the area of $\triangle ABC$.

Problem 10:
Let P be the set of permutations $a_1, a_2, a_3, \ldots, a_8$ of $1, 2, 3, \ldots, 8$ such that $|a_i - i| \leq 1$ for each i. The expected value of $\sum_{i=1}^{8} i a_i$ over a permutation chosen uniformly at random from P is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the remainder when $m + n$ is divided by 1000.

Problem 11:
In convex quadrilateral $ABCD$, $\angle ADB$ and $\angle BCD$ are right, and $AB = AC = 6$. Suppose that $\angle BAC$ is maximized subject to these conditions. Find the square of the area of $ABCD$.

Problem 12:
Define $f(x) = 2x - x|x|$ for $-2 \leq x < 2$ and $f(x + 4) = f(x)$ for all real numbers x. Find the number of real numbers x with $-30 \leq x \leq 30$ for which $f(x) = f(x^2)$.

Problem 13:
In trapezoid $ABCD$ with $AB = 7 > CD$, $AD = 8$, and $\angle B$ and $\angle C$ right, a point E lies outside of $ABCD$ such that $\angle BAE$ and $\angle DBE$ are right and quadrilateral $ADCE$ is cyclic. Then $BD = m\sqrt{n} - p$, where m, n, and p are positive integers, and n is not divisible by the square of any prime. Find $m + n + p$.

Problem 14:
For a positive integer n relatively prime to 22, let $p(n)$ denote the length of the repeating block in the base-22 expansion of $\frac{1}{n}$. Find the least positive integer N for which

$$\sum_{a=1}^{N} \sum_{b=1}^{N} p(3^a \cdot 7^b)$$

is divisible by 2025.

Problem 15:
In convex quadrilateral $ABCD$ with $\angle ACB$ and $\angle ADB$ right, let P and Q be on sides $\overline{AD}$ and $\overline{BC}$, respectively, such that $AP = 49$, $DP = 14$, and $CQ = 60$. The circumcircles of $\triangle ACP$ and $\triangle BDQ$ intersect at distinct points X and Y. Suppose that line XY bisects $\overline{AB}$. Find BC.